

Co-Pilot for Lawyers

An AI assistant that reads legal documents, extracts data, flags rule deviations, and answers questions with auditable source citations.

Group 88: Konstantin Unterweger, Vincent A. Balau, David Fellner, Florian Ebner | Applied Generative AI and LLM-based Systems · 2026

Abstract

Lawyers spend disproportionate time reading contracts and regulatory documents.

Our system covers:

- **Extraction:** pulls user defined key facts (parties, dates, governing law,...) out of source documents
- **Rule-based Review:** flags every passage that breaks/fulfils a predefined rule
- **Cross-document Q&A:** answers questions, with a precise reference to the source passage

Initial Situation & Problem

Three recurring pain points for legal professionals:

- **Multi-document extraction:** manually pulling key terms, dates, and clauses from dozens of contracts.
- **Risk review:** checking compliance against internal policies and regulatory frameworks.
- **Cross-document research:** finding answers that span multiple documents without missing context.

The screenshot shows the LEGAL CO-PILOT interface. On the left, a sidebar lists three main modes: 1 Extraction, 2 Rule Review, and 3 Q&A (highlighted). Below the modes is a 'DOCUMENTS' section listing several legal documents, with a '+ Upload document' button at the bottom. The main area is titled 'Q&A' and includes the instruction 'Ask a question across all documents — answer with exact citations'. A dropdown menu at the top right of the main area is set to 'Corpus: user docs'. A user question is entered: 'Wie lange gilt die Probezeit?'. The AI response states: 'Die Probezeit beträgt die ersten drei Monate des Arbeitsverhältnisses. Während dieser Zeit kann das Arbeitsverhältnis von beiden Seiten jederzeit ohne Angabe von Gründen und ohne Einhaltung einer Frist aufgelöst werden [DOC: 02_Arbeitsvertrag_Senior_Developer.txt, p. 1, clause 2.2]'. Below the response, the source is cited as '02_Arbeitsvertrag_Senior_Developer.txt · clause 2.2'. At the bottom, a chat field contains the example question: 'e.g. In which contracts can the landlord terminate without cause?'. Red dashed boxes and arrows highlight key features: the three main modes, the uploaded user documents list, the context selection dropdown, the AI answer with its citation, and the chat input field.

LEGAL CO-PILOT

Q&A
Ask a question across all documents — answer with exact citations

Corpus: user docs

3 Main Modes

1 Extraction
2 Rule Review
3 Q&A

DOCUMENTS

- Dienstvertrag_Pflege_Graz.pdf [u...
- Dienstvertrag_Bank_Salzburg.pdf...
- Arbeitsvertrag_Maschinenbau_Li...
- Arbeitsvertrag_Marketing_Wien....
- 02_Arbeitsvertrag_Senior_Devel...

+ Upload document

Context selection (user documents/ legal texts/ both)

Model answers with references to uploaded documents

Chat field, to ask any questions about uploaded documents

U: Wie lange gilt die Probezeit?

AI: Die Probezeit beträgt die ersten drei Monate des Arbeitsverhältnisses. Während dieser Zeit kann das Arbeitsverhältnis von beiden Seiten jederzeit ohne Angabe von Gründen und ohne Einhaltung einer Frist aufgelöst werden [DOC: 02_Arbeitsvertrag_Senior_Developer.txt, p. 1, clause 2.2].

02_Arbeitsvertrag_Senior_Developer.txt · clause 2.2

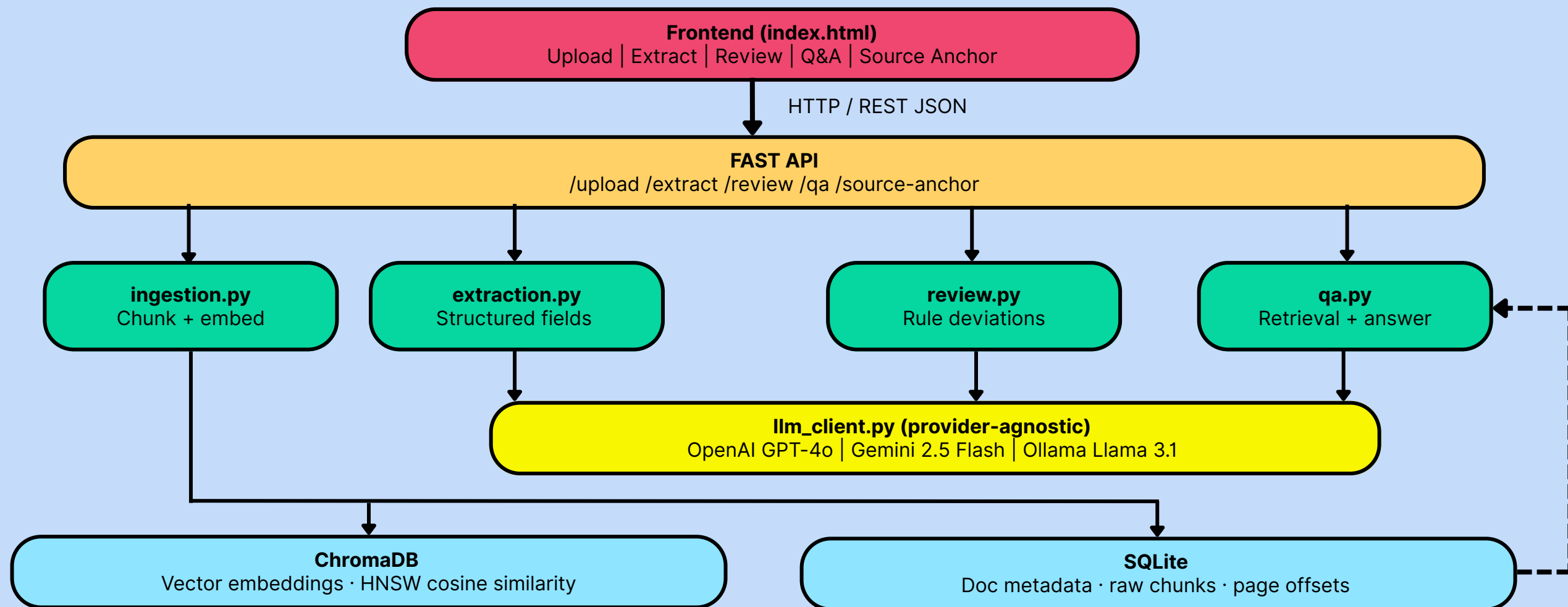
e.g. In which contracts can the landlord terminate without cause?

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Architecture



Key Design Decisions

- 1 Dual Storage: Vector + Relational**
 - ChromaDB for semantic search (HNSW cosine)
 - SQLite stores metadata, raw chunks, and page offsets
- 2 Chunk Based Retrieval**
 - ~1500-char chunks with 200-char overlap, each carrying page/offset info for citation; only relevant chunks sent to LLM.
- 3 RAG Instead of Fine-Tuning**
 - each query retrieves chunks fresh from database and adds it to LLM context
 - new documents immediately available without any training effort

Metrics

- 1 ~Extract Quote**
 - Extracted fields with supporting quotes
- 2 ~Extract Page**
 - Extracted fields with page references
- 3 ~Review Page**
 - Review findings with supporting quotes
- 4 ~Review Page**
 - Review findings with page references
- 5 ~QA w/ Sources**
 - Answers containing source references
- 6 ~Avg Sources**
 - Average number of sources per answer

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LEGAL CO-PILOT

1 Extraction

2 Rule Review

3 Q&A

Mode selection

Document selection

DOCUMENTS

Dienstvertrag_Pflege_Graz.pdf [u...

Dienstvertrag_Bank_Salzburg.pdf...

Arbeitsvertrag_Maschinenbau_Li...

Arbeitsvertrag_Marketing_Wien...

O2_Arbeitsvertrag_Senior_Devel...

+ Upload document

Extraction

Define fields — get a structured table with exact source links

corpus selection dropdown menu

Corpus: user docs

Run ↗

Button to run the extraction feature

Results — 5 documents × 5 fields

← Edit

Button to edit the fields for extraction

user defined fields

specified and successfully identified field in the legal text

Result row for each Document

field not specified in legal document

Document	Berufsbezeichnung	Startdatum	Monatsgehalt	Bonusregelung	Arbeitszeit (Stunden pro Woche)
Dienstvertrag_Pfleg...	diplomierte Gesundheits- und Krankenpflegerin (DGKP) clause 3.1 found	1. Juni 2025 § 2.1 found	EUR 3.420,-- § 5.1 found	— p. 2 not_specified	37 § 4.1 found
Dienstvertrag_Bank_...	Head of Credit Risk Management § 3.1 found	1. September 2025 § 2.1 found	10.142,86 Page 2, clause 5.1 found	variable Vergütungssystem Page 2, paragraph 5.2 found	38,5 § 4 Abs. 1 found
Arbeitsvertrag_Masc...	Senior Konstrukteur § 3.1 found	1. Mai 2025 § 2.1 found	EUR 4.850,-- § 5.1 found	Projektprämie 5.3 found	38,5 § 4.1 found
Arbeitsvertrag_Mark...	Junior Content & Social Media Manager § 3.1 found	1. März 2025 § 2.1 found	EUR 2.650,-- § 5.1 found	— p. 2 not_specified	38,5 Stunden § 4.1 found
02_Arbeitsvertrag_S...	Senior Software Developer § 3.1	1. Jänner 2025 §	EUR 6.800,-- § 5.1	15% des Jahresbruttogehalts	38,5 § 4 Abs. 1 found

User Workflow (Extraction Feature)



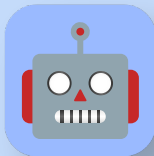
1

- Upload contracts or other law documents as pdfs



2

- User types fields to extract
- e.g.: "Berufsbezeichnung", "Monatsgehalt", ...



3

- LLM extracts fields
- marks if document references topic



4

- tabular view over all extracted features of uploaded documents
- citations referenced if found